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Compressible Flow

Compressible Flow Lab Report

Calibration of Supersonic Wind Tunnel Using Pressure Data & Schlieren System

Data collected on date: 03/25/2026
Due date: 04/12/2026

Introduction and Objectives

Supersonic wind tunnels are essential tools in aerospace engineering for studying high-speed compressible flows and validating aerodynamic theory. Unlike subsonic flows, supersonic flow fields are characterized by strong nonlinear phenomena such as shock waves, expansion fans, and large variations in pressure, temperature, and density. These effects play a critical role in the design of high-speed aircraft, rockets, and propulsion systems.

The Syracuse University supersonic wind tunnel is a blow-down facility that generates high-speed flow by releasing compressed air through a converging-diverging nozzle. The nozzle is designed to produce a test section flow with a Mach number of approximately 2. This type of facility allows for controlled experimentation and comparison between theoretical predictions and measured flow behavior.

In this experiment, both pressure measurements and Schlieren imaging were used to analyze the flow. Pressure data provides quantitative information about flow properties such as Mach number, while Schlieren visualization enables qualitative observation of density gradients, allowing shock structures to be identified and measured. The combination of these methods allows for a comprehensive evaluation of the wind tunnel performance.

The primary objective of this experiment was to calibrate and evaluate the performance of a blow-down supersonic wind tunnel operating near Mach 2 using both pressure-based methods and optical flow visualization.

The specific objectives of this experiment were to determine the test section Mach number using multiple independent methods, including isentropic pressure relations, the Rayleigh Pitot tube relation, and oblique shock theory based on Schlieren imaging. In addition, pressure data collected from transducers located in the stagnation chamber, test section, and model surfaces was analyzed to characterize the flow. Schlieren images were used to identify and measure shock structures, including the shock angle generated by a ramp model. The experiment also aimed to apply compressible flow theory to the measured data and compare the results with theoretical expectations. Furthermore, uncertainty in the calculated Mach numbers was quantified using error propagation methods based on the known accuracy of the sensors. Finally, the overall performance of the wind tunnel was assessed by comparing the experimental results to the expected design Mach number.

Experimental Facility and Instrumentation

The experiment was conducted using the Syracuse University blow-down supersonic wind tunnel located within an anechoic chamber at the Skytop Laboratory. The wind tunnel operates by releasing compressed air from a high-pressure reservoir into a stagnation chamber, where the flow is directed through a converging-diverging (CD) nozzle to achieve supersonic conditions in the test section. The nozzle is designed to produce a nominal test section Mach number of approximately 2. Downstream of the nozzle, the flow enters a constant-area test section where interchangeable models can be installed. In this experiment, two model inserts were used: a 10° ramp to generate an oblique shock and a Pitot tube to produce a normal shock. These configurations allow for both pressure-based and optical methods of determining the Mach number.

The facility is equipped with three pressure transducers used to measure flow properties. Transducer 1 measures the stagnation pressure in the reservoir, while Transducer 2 records the static pressure in the test section. Transducer 3 measures either the ramp surface static pressure or the Pitot stagnation pressure depending on the model configuration. Each transducer has a specified measurement range and accuracy, which are used in subsequent uncertainty analysis. Pressure data is acquired using a data acquisition system connected to the transducers, allowing time-resolved measurement of pressure during each wind tunnel run. In addition to pressure measurements, the test section is equipped with a Schlieren optical system. This system uses a high-intensity light source and mirrors to visualize density gradients in the flow, enabling direct observation of shock waves and other compressible flow features. The anechoic chamber surrounding the facility minimizes acoustic reflections and environmental disturbances for testing that is not used in this experimentation, but it works to reduce external noise interference and echo during testing in supersonic simulations.

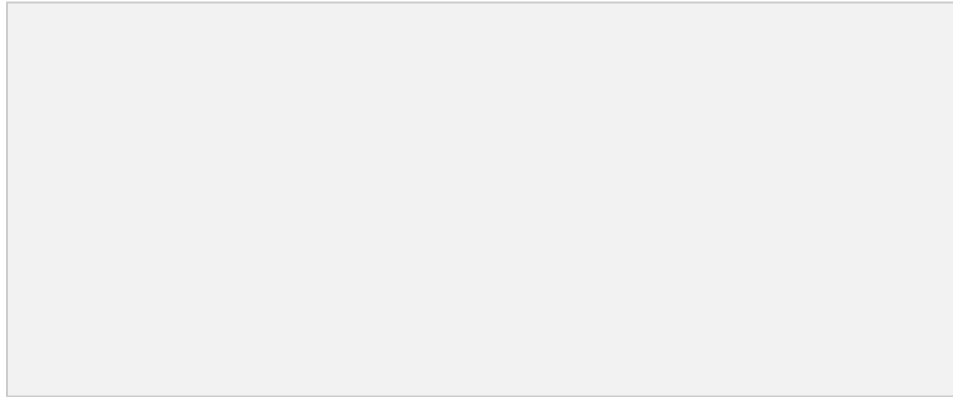


Figure 1: Supersonic blow-down wind tunnel located in the anechoic chamber at Syracuse University Skytop Laboratory. The system includes the converging-diverging nozzle, test section, and downstream diffuser, along with mounting structures and instrumentation used for pressure measurement and Schlieren visualization

Methods

The objective of this experiment is to determine the Mach number in the test section of a supersonic wind tunnel using both pressure-based measurements and optical flow visualization. The performance parameter of interest, the Mach number, is calculated using three independent methods: isentropic flow relations, the Rayleigh Pitot tube relation, and oblique shock theory.

Each method relies on different measurable quantities and physical assumptions. The isentropic method assumes inviscid, adiabatic flow with no shocks between the reservoir and test section. The Rayleigh Pitot method accounts for the presence of a normal shock ahead of the Pitot tube. The oblique shock method uses geometric measurements from Schlieren images to determine the upstream Mach number from shock angle relations.

By comparing Mach number estimates obtained from these independent approaches, the performance of the wind tunnel can be evaluated and validated against the expected design condition.

Isentropic relation is the first of the governing equations used for this analysis. For isentropic flow, the Mach number is determined from the stagnation to static pressure ratio:

$$M_1 = \sqrt{\frac{2}{\gamma-1} \left[\left(\frac{P_0}{p_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]}$$

Where:

- P_0 is the stagnation pressure.
- p_1 is the test section static pressure.
- $\gamma = 1.4$ for air

This method assumes that the flow is isentropic between the tank reservoir and the test section. Under this assumption, the stagnation pressure in the test section, P_{01} , is equal to the reservoir stagnation pressure, P_0 , which is directly measured during the experiment.

The Rayleigh pitot tube relation states that when a Pitot tube is placed in supersonic flow, a normal shock forms ahead of the probe. The Mach number is determined from the ratio of stagnation pressure behind the shock P_{02} to the upstream static pressure p_1 :

$$\frac{P_{02}}{p_1} = \left[\frac{(\gamma+1)^2 M^2}{4\gamma M^2 - 2(\gamma-1)} \right]^{\frac{\gamma}{\gamma-1}} * \frac{(1-\gamma) + 2\gamma M^2}{\gamma+1}$$

This equation is implicit in Mach number and is solved numerically using a root-finding method.

For the ramp model, the oblique shock relation states that an oblique shock forms at the leading edge. The Mach number is determined using the $\theta - \beta - M$ relation:

$$\tan \theta = 2 \cot \beta * \frac{M^2 \sin^2 \beta - 1}{M^2 (\gamma + \cos 2\beta) + 2}$$

Where:

- $\theta = 10^\circ$ is the ramp angle
- β is the measured shock angle from Schlieren images.

This equation is also solved numerically for Mach number.

Pressure measurements were obtained using three transducers installed in the wind tunnel:

- Transducer 1: stagnation pressure P_0
- Transducer 2: test section static pressure p_1
- Transducer 3:
 - ramp surface static pressure (ramp case), or
 - Pitot stagnation pressure P_{02} (Pitot case)

The pressure data was recorded as a function of sample number using a data acquisition system. For each run, a steady-state region was identified from the pressure-time plots, and the mean pressure values were computed over this region.

All pressure values were converted from psig to psia by adding atmospheric pressure, 14.7 psi before being used in the governing equations.

In addition to pressure measurements, Schlieren imaging was used to visualize the flow field. The shock angle, β , was measured directly from the images using an online image angle measurement tool and used in the oblique shock analysis.

The accuracy of the pressure transducers is specified as:

- Transducer 1: ± 1.0 psi
- Transducer 2: ± 0.5 psi for $|p| < 15$ psig
- Transducer 3:
 - ± 0.5 psi for $|p| < 15$ psig
 - ± 0.06 psi for $|p| > 15$ psig

These uncertainties were used in the propagation of error for Mach number calculations.

Uncertainty in the calculated Mach number was determined using the root-sum-square (RSS) error propagation method:

$$u_M = \sqrt{\left(\frac{\partial M}{\partial x_1} u_{x_1}\right)^2 + \left(\frac{\partial M}{\partial x_2} u_{x_2}\right)^2}$$

where x_1 and x_2 represent measured variables such as pressure or shock angle.

For the isentropic method, uncertainty was propagated from P_0 and p_1 using numerical differentiation. For the Rayleigh Pitot method, the Mach number was computed numerically for slightly perturbed values of p_1 and P_{02} , and central difference approximations were used to estimate sensitivity to each variable. For the oblique shock method, uncertainty was estimated by perturbing the measured shock angle β and evaluating the corresponding change in Mach number. A measurement uncertainty of $\pm 1^\circ$ was assumed for the shock angle. This approach provides a quantitative estimate of the confidence in the calculated Mach numbers based on measurement limitations.

Experimental Design

The experiment was conducted using a blow-down supersonic wind tunnel operating with compressed air as the working fluid. The flow was expanded through a converging-diverging nozzle designed to achieve a nominal test section Mach number of approximately 2. The facility operated under transient conditions, where the pressure in the reservoir decreased over time during each run.

Data was collected during the period in which the flow reached a quasi-steady condition. This steady region was identified from the pressure measurements, and only this portion of the data was used for analysis.

The experiment was conducted through a series of wind tunnel runs using different test configurations to evaluate the Mach number using multiple methods. Prior to testing, the wind tunnel was pressurized using compressed air, and all instrumentation, including pressure transducers and the Schlieren system, was initialized and calibrated. In the first configuration, a 10° ramp model was installed in the test section to generate an oblique shock. The tunnel was operated, and pressure data was recorded from all three transducers while Schlieren images were captured simultaneously to visualize the shock structure. These images were later used to measure the shock angle. In the second configuration, the ramp model was replaced with a Pitot tube probe. The wind tunnel was again operated under similar

conditions, and pressure data was collected. In this case, the Pitot tube generated a normal shock ahead of the probe, allowing the Rayleigh Pitot relation to be applied.

For each run, pressure data was recorded continuously, and a steady-state region was identified from the pressure traces. Average values of stagnation and static pressures were computed over this region and used to calculate Mach number using the different analytical methods. For the ramp configuration, the shock angle was measured from Schlieren images by determining the angle between the freestream direction and the shock wave, and this measurement was used in the oblique shock relation to compute Mach number. Two primary runs were conducted: a ramp run, used to evaluate Mach number using isentropic relations and oblique shock theory, and a Pitot run, used to evaluate Mach number using isentropic relations and the Rayleigh Pitot method. These runs provided independent estimates of Mach number based on different physical principles, allowing for comparison and validation of the wind tunnel performance.

Several limitations affect the accuracy and reliability of the experimental results. The blow-down nature of the wind tunnel results in transient flow conditions, requiring careful selection of a steady-state region for analysis. Measurement uncertainty in the pressure transducers propagates into the calculated Mach number. Additionally, the shock angle measured from Schlieren images is subject to human error and limitations in image resolution, which affects the accuracy of the oblique shock method. The analysis also relies on simplifying assumptions, such as isentropic flow and ideal shock behavior, while real flow effects including boundary layers, viscous losses, and flow non-uniformity may introduce discrepancies. Finally, only two configurations were tested, limiting the ability to assess repeatability and variability in the results.

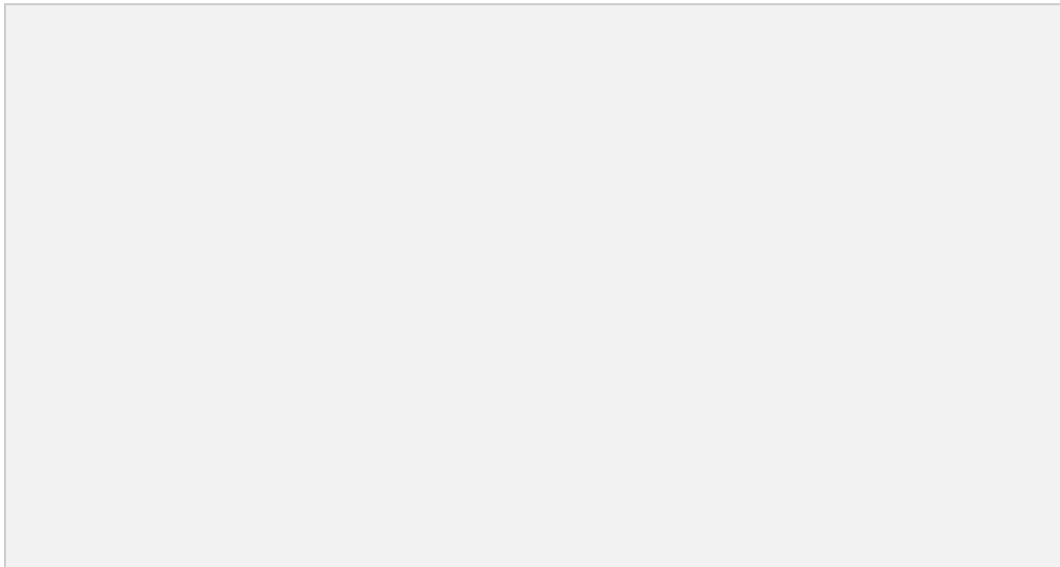


Figure 3: Schlieren image of the Pitot tube configuration showing the bow shock formed ahead of the probe. The standoff distance between the shock and the probe tip is indicated, illustrating the normal shock structure in supersonic flow.

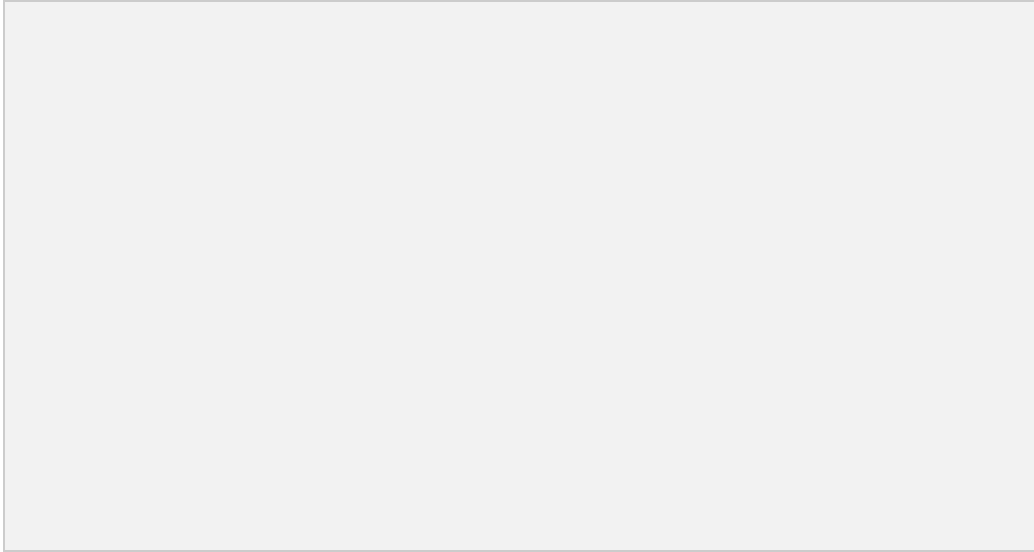


Figure 4: Schlieren image of the ramp configuration showing the oblique shock generated by a 10° ramp. The measured shock angle, $\beta = 37^\circ$, is indicated along with the ramp angle and a horizontal reference line used for angle determination.

Results and Discussions

Pressure data from both the ramp and Pitot tube runs were first plotted as a function of sample number to identify a steady operating region. In both cases, the pressure traces showed a clear startup transient, a quasi-steady period, and then a gradual decay as the blow-down tunnel reservoir pressure decreased over time. This behavior is consistent with the expected operation of a blow-down supersonic wind tunnel, where stable conditions can only be maintained over a limited portion of the run. The provided lab handout also notes that the pressure data takes time to converge and that this must be considered when selecting data for analysis.

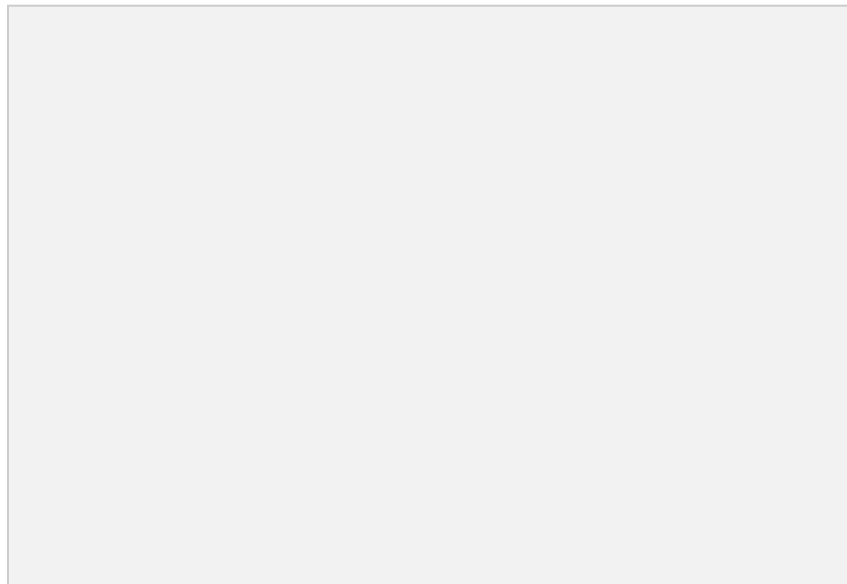


Figure 4: Pitot tube configuration pressure data plotted as a function of sample number. The plot includes stagnation pressure, test section static pressure, and Pitot stagnation pressure measured during the blow-down run, illustrating the formation of a steady region used for Mach number analysis.

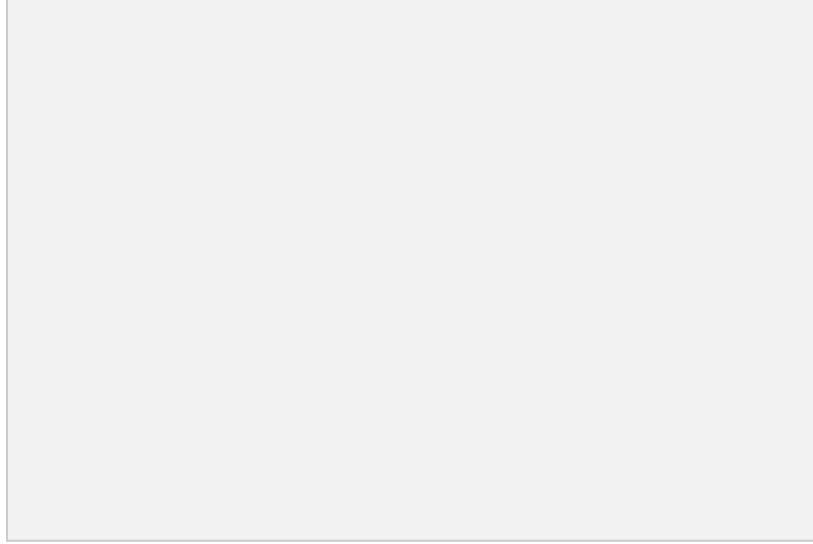


Figure 5: Ramp configuration pressure data plotted as a function of sample number. The plot includes stagnation pressure, test section static pressure, and ramp surface static pressure measured during the blow-down run, showing the startup transient, quasi-steady region, and gradual pressure decay.

For the ramp run, the stagnation pressure increased to a peak value of roughly 40 psig while the test section static pressure and ramp surface static pressure remained negative relative to atmospheric pressure, indicating accelerated flow in the test section. The ramp surface pressure was consistently higher than the test section static pressure, which is physically reasonable because the oblique shock generated by the ramp causes a pressure rise across the shock. In the Pitot run, the stagnation pressure again rose to a similar level, while the Pitot stagnation pressure measured by Transducer 3 remained significantly higher than the test section static pressure. This is expected because the Pitot probe measures the stagnation pressure behind the normal shock formed ahead of the probe, while the static port measures the lower freestream static pressure. The transducer roles and model configurations are consistent with the facility description in the handout.

Using the steady-state average pressures, Mach number was calculated by four methods. The isentropic method gave $M = 1.979 \pm 0.047$ for the ramp run and $M = 1.975 \pm 0.043$ for the Pitot run. The Rayleigh Pitot method gave $M = 1.960 \pm 0.070$, and the oblique shock method using the measured ramp shock angle of $\beta = 37^\circ$ gave $M = 2.125 \pm 0.059$. These results are summarized in the bar chart with error bars.

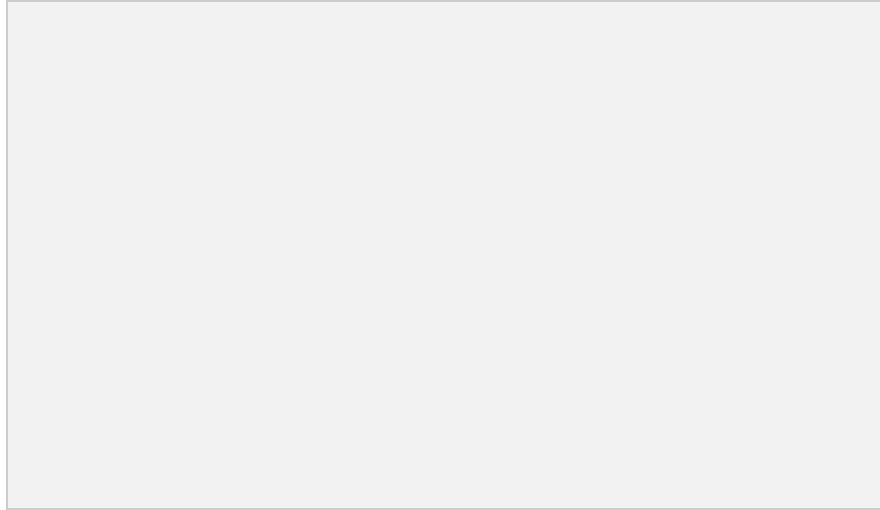


Figure 6: Mach number estimates with uncertainty obtained using isentropic relations, Rayleigh Pitot theory, and oblique shock analysis. Error bars represent propagated measurement uncertainty, and the dashed line indicates the design Mach number of 2 for the wind tunnel.

Overall, the pressure-based Mach numbers are in very good agreement with one another and are all close to the expected design Mach number of 2. The handout states that the nozzle is designed to achieve a test section Mach number of approximately 2, so values near this level indicate that the tunnel was operating as intended. The two isentropic results are nearly identical, which suggests that the selected steady-state regions were appropriate and that the reservoir and test section pressure measurements were internally consistent across the two runs. The Rayleigh Pitot result is slightly lower than the isentropic values, which is also reasonable. Unlike the isentropic method, the Rayleigh Pitot method explicitly accounts for the normal shock ahead of the probe, and it is more sensitive to both pressure ratio measurement error and non-ideal losses in the flow. The handout specifically notes that this method requires numerical treatment and sensitivity analysis because the relation is implicit in Mach number.

The oblique shock result of $M = 2.125$ is somewhat higher than the pressure-based results, but it is still reasonably close and within the general range expected for a Mach 2 facility. This difference is likely due to uncertainty in measuring the shock angle from the Schlieren image. Even a small change in the measured value of β can noticeably affect the Mach number obtained from the $\theta - \beta - M$ relation. In addition, the shock in the image is not perfectly sharp over its full length, and the exact point at which the line should be drawn is somewhat subjective. This makes the optical method more sensitive to user interpretation than the pressure-based methods, even though it remains very useful for visual confirmation of the flow structure. The handout identifies the ramp as a 10° wedge and states that the shock angle from Schlieren images can be used with oblique shock theory to determine Mach number.

The uncertainty trends also make physical sense. The two isentropic methods produced the smallest uncertainties, since they depend on direct pressure measurements and a relatively simple explicit equation. The Rayleigh Pitot method produced a somewhat larger uncertainty because it depends on two measured pressures through an implicit nonlinear relation. The oblique shock result also has noticeable uncertainty because it depends on the measured shock angle, which is less precise than the pressure transducer readings. The pressure transducer accuracies given in the handout support this analysis, and the uncertainty propagation approach follows the error propagation framework described for the lab.

Several additional observations can be made from the experiment. First, stable flow conditions were not instantaneous; both pressure plots show a clear ramp-up period before the quasi-steady region was reached. Second, because the facility is a blow-down tunnel, the stagnation pressure gradually decayed during each run, making the duration of useful steady data limited. Third, the Pitot pressure trace showed more noticeable fluctuations than the ramp surface pressure trace, likely because the probe and bow shock configuration is more sensitive to unsteadiness in the flow. Finally, small variations in optical clarity and line placement in the Schlieren images likely contributed to the discrepancy between the oblique shock estimate and the pressure-based values.

Taken together, the results are consistent with compressible flow theory and support the conclusion that the tunnel was producing supersonic flow close to its intended Mach 2 operating condition. The close agreement among the pressure-based methods provides strong evidence that the wind tunnel calibration was successful, while the Schlieren-based result offers an independent visual confirmation of the supersonic flow field and associated shock structure.

Conclusions and Recommendations

The results of this experiment demonstrate that the Syracuse University supersonic wind tunnel is capable of producing flow conditions close to its intended design Mach number of 2. Mach numbers calculated using multiple independent methods were in good agreement, with the isentropic methods yielding values of approximately 1.98, the Rayleigh Pitot method yielding 1.96, and the oblique shock method yielding 2.13. These values are all reasonably close to Mach 2 and consistent with theoretical expectations for the facility.

The agreement between the two isentropic results indicates that the pressure measurements were reliable and that the steady-state regions selected for analysis were appropriate. The slightly lower Mach number obtained from the Rayleigh Pitot method is consistent with the presence of a normal shock and the increased sensitivity of this method to pressure measurement uncertainty and non-ideal flow effects. The higher Mach number obtained from the oblique shock method is attributed primarily to uncertainty in measuring the shock angle from Schlieren images.

Overall, the results are consistent with fundamental compressible flow theory. The pressure-based methods provided quantitatively consistent estimates of Mach number, while the Schlieren imaging provided qualitative confirmation of shock structures and supersonic flow behavior. The close agreement between methods indicates that the experimental objectives were successfully achieved.

Although the experiment produced reliable results, several improvements could enhance accuracy and repeatability. The oblique shock method is sensitive to the measured shock angle, so using higher-resolution Schlieren images, digital image processing, or a more accurate method could reduce user error and improve accuracy. The pressure data was collected during transient blow-down operation, which limits the duration of steady conditions; implementing a method to better regulate or extend steady-state operation would improve data consistency and allow for more precise averaging. In addition, only one run was performed for each configuration, so repeating the experiment multiple times would enable statistical analysis of variability and improve confidence in the results. While the pressure transducers provided reasonable accuracy, more frequent calibration or the use of higher-precision sensors could further reduce uncertainty in pressure measurements and improve Mach number estimates. The analysis also assumes ideal compressible flow, and incorporating corrections for viscous effects, boundary layer growth, and flow non-uniformity could improve agreement between theoretical

predictions and experimental results. Finally, the steady-state region was identified visually from the pressure plots, and implementing an automated method for detecting steady-state conditions could reduce subjectivity and improve consistency in data processing.

References

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Appendix

Raw pressure data from both the ramp and Pitot tube configurations were recorded using a data acquisition system and stored in CSV format (Ramp_Data.csv and Pitot_Data.csv). These datasets include time-resolved measurements from all three pressure transducers and were used to generate the pressure versus sample number plots shown in the Results section. Additional plots of the full pressure traces are included to illustrate the identification of startup transients, quasi-steady regions, and the decay phase associated with blow-down operation.

The instrumentation used in this experiment consisted of three pressure transducers with specified accuracy ranges. Transducer 1 measured stagnation pressure in the reservoir, Transducer 2 measured test section static pressure, and Transducer 3 measured either ramp surface static pressure or Pitot stagnation pressure depending on the configuration. The transducer accuracies and operating ranges were incorporated into the uncertainty analysis presented in the Methods section.

All data processing, Mach number calculations, and uncertainty analysis were performed using MATLAB. The analysis code includes procedures for selecting steady-state regions, computing average pressure values, solving the governing equations for Mach number using numerical methods, and propagating uncertainty using central difference approximations and root-sum-square error analysis. The MATLAB scripts used in this analysis are included in the appendix to ensure reproducibility of results.

Schlieren images used for flow visualization are also included, showing both the oblique shock generated by the ramp and the bow shock formed ahead of the Pitot tube. These images have been annotated to indicate key measurements such as shock angle and standoff distance, which were used in the corresponding analytical methods.